Statistical Modelling of Factors Affecting Student Academic Performance

Oluwadurotimi Alade

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1 Introduction

1.1 Background of the Study

Student academic performance is influenced by a combination of academic, personal, and environmental factors. Understanding these factors can help educators identify students who may require additional support and develop interventions that improve learning outcomes.

Statistical modelling provides a useful framework for studying these relationships. By analysing measurable characteristics such as study time, attendance, previous academic performance, and access to learning resources, it is possible to investigate how different factors are associated with students' academic outcomes.

1.2 Problem Statement

Educational institutions often collect information about students, but the relationships between different characteristics of students and academic performance may not be immediately apparent. A statistical model can help identify important predictors of performance and quantify their relationships with academic outcomes.

1.3 Aim of the Study

The purpose of this study is to develop a statistical model to examine factors associated with student academic performance.

1.4 Research Objectives

The specific objectives are to:

1. identify factors that are associated with academic performance;
2. examine the relationship between study behavior and academic outcomes;
3. develop a statistical model for predicting academic performance; and
4. evaluate the performance of the proposed model.

1.5 Research Questions

The study addresses the following questions:

1. What factors are significantly associated with academic performance of the student?
2. How does study time relate to academic performance?
3. Can academic performance be predicted using measurable student characteristics?

1.6 Significance of the Study

The findings of this study may be useful to educators, academic administrators, and other stakeholders interested in improving student academic outcomes. By identifying factors that are associated with academic performance, the study may provide insights that can support evidence-based academic interventions and student support strategies.

The statistical model developed in this study may also provide a framework for examining student performance using measurable characteristics. This can contribute to a better understanding of the factors associated with differences in academic outcomes and provide a basis for further research.

1.7 Scope of the Study

This study focuses on measurable factors that may be associated with student academic performance. The analysis will consider selected student characteristics, including study time, attendance, previous academic performance, and access to learning resources, depending on the availability and suitability of the data.

The study will use statistical modelling techniques to examine the relationships between the selected factors and academic performance. The scope is limited to the variables available in the study dataset and does not attempt to account for every possible factor that may influence students' academic outcomes.

2 Literature Review

2.1 Academic Performance

Academic performance is commonly used as an indicator of students' educational outcomes. It may be measured using examination scores, continuous assessment results, grade point averages, or other academic indicators (Al-Tameemi et al., 2023).

2.2 Factors Associated with Academic Performance

Several factors may influence students' academic outcomes. These factors can be broadly classified into academic, personal, social, and demographic characteristics (Al-Tameemi et al., 2023). Study habits, study skills, and attitudes towards learning have also been identified as important predictors of academic performance (Credé and Kuncel, 2008).

These factors may interact with one another and may operate through different aspects of students' academic and personal experiences. For example, time management, attendance, sleep, and self-esteem have been examined together as factors that may be associated with academic performance among university students (Masnan et al., 2025). Considering these factors together provides a broader basis for understanding academic performance than focusing on a single characteristic in isolation.

2.3 Theoretical Framework: Maslow's Hierarchy of Needs

The range of personal and environmental factors associated with academic performance suggests that students' educational outcomes cannot always be considered independently of their broader needs and circumstances. Maslow's Hierarchy of Needs provides a theoretical perspective for considering how different human needs may influence motivation and behaviour (Freitas and Leonard, 2011). The theory proposes five broad categories of needs: physiological needs, safety needs, love and belonging, esteem, and self-actualisation. In an educational context, these needs provide a useful framework for considering factors that may influence students' ability to engage effectively in learning (Freitas and Leonard, 2011).

Physiological needs include basic requirements such as adequate food, rest, and physical wellbeing. Safety needs relate to security and freedom from conditions that may cause significant physical or psychological distress. Love and belonging involve relationships with family, peers, and other members of the learning community, while esteem relates to confidence, recognition, and a sense of competence. Self-actualisation represents the pursuit of personal growth and the achievement of one's potential.

The application of Maslow's framework to education suggests that students' academic experiences may be influenced by conditions extending beyond the classroom. These conditions may affect students' motivation, engagement, and

ability to participate effectively in learning. Thus, the framework provides a useful perspective for considering the broader personal and environmental factors associated with academic performance (Freitas and Leonard, 2011).

In this study, Maslow's Hierarchy of Needs is used as a theoretical lens rather than as a direct statistical model. It provides a broader perspective for understanding how personal and environmental conditions may relate to academic performance, while the statistical analysis will focus on measurable variables available in the study dataset.

2.4 Study Time and Performance

Study time represents the amount of time a student spends engaging with academic materials outside scheduled classes. Although study time is an important aspect of students' academic behaviour, research suggests that the amount of time spent studying does not necessarily translate directly into higher academic performance. The effectiveness and quality of study practices may be important considerations alongside the amount of time spent studying (Plant et al., 2005).

Study time should also be considered alongside the strategies students use during learning. The effectiveness of learning may depend not only on the amount of time devoted to academic activities but also on how that time is used. Research examining classroom attendance and learning strategies suggests that both behavioural and learning-related factors can contribute to differences in learning outcomes (Tetteh, 2018). This supports the consideration of study behaviour as a broader construct when examining factors associated with academic performance.

2.5 Attendance and Performance

Class attendance is another measurable factor that may be associated with academic performance. Research has examined the relationship between attendance and academic outcomes, with findings suggesting that attendance can be an important predictor of student performance (Arnold, 2018). The timing and pattern of student absences may also influence the relationship between absenteeism and academic achievement (Keppens, 2023).

The relationship between attendance and academic performance may also depend on the learning context. Evidence from synchronous university courses indicates that class attendance can be associated with academic performance, suggesting that participation in scheduled learning activities may be relevant to students' academic outcomes (Ha et al., 2024). These findings further support the inclusion of attendance as a measurable factor when examining student academic performance.

2.6 Statistical Modelling

The discussion of study behaviour, attendance, and other factors associated with academic performance highlights the need for an approach that can exam-

ine several explanatory variables simultaneously. Statistical modelling provides such a framework by allowing relationships between a dependent variable and one or more explanatory variables to be examined systematically. In this study, multiple linear regression is considered as a suitable approach for modelling academic performance as a function of measurable student characteristics.

A general multiple linear regression model can be expressed as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon,$$

where Y represents academic performance, $X_1, \ldots, X_p$ represent explanatory variables, β_0 is the intercept, $\beta_1, \ldots, \beta_p$ are regression coefficients, and ε is the error term. Previous research has demonstrated the use of multiple linear regression and principal component analysis for predicting students' academic performance (Yang et al., 2018).

3 Methodology

3.1 Research Design

This study adopts a quantitative research design to examine factors associated with student academic performance. Quantitative research involves examining measurable variables using numerical data and statistical procedures to investigate relationships between variables and test research questions (Creswell and Creswell, 2023).

The study focuses on measurable characteristics of students and uses statistical modelling to investigate the relationship between these characteristics and academic outcomes. This approach is appropriate for the study because student academic performance and the explanatory factors considered can be represented using numerical measures and analysed statistically.

3.2 Study Variables

The dependent variable in the study is student academic performance. The explanatory variables represent measurable characteristics that may be associated with academic performance, including study time, attendance, previous academic performance, and access to learning resources. Previous research has identified study behaviour and attendance among factors that may be associated with students' academic outcomes (Masnan et al., 2025; Ha et al., 2024).

The variables are considered jointly so that the statistical model can examine the relationship between academic performance and several explanatory factors simultaneously. The dependent variable represents the outcome of interest, while the explanatory variables are included to investigate their individual contributions to variations in academic performance.

For the purpose of the regression analysis, the variables will be represented numerically where appropriate. This allows the observed student-level data to be incorporated into the statistical model and provides a basis for estimating the relationship between academic performance and the selected explanatory variables.

The variables are considered jointly so that the statistical model can examine the relationship between academic performance and several explanatory factors simultaneously.

3.3 Multiple Linear Regression Model

Since the study considers several explanatory variables simultaneously, multiple linear regression is used to examine their relationship with student academic performance. This method allows the contribution of each explanatory variable to be estimated while holding the other variables constant (Yang et al., 2018).

The general form of the multiple linear regression model is given by:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_p X_{pi} + \varepsilon_i \tag{1}$$

where:

$$
\begin{aligned}
Y_i &= \text{academic performance of student } i, \\
\beta_0 &= \text{intercept of the regression model}, \\
\beta_j &= \text{coefficient associated with explanatory variable } X_j, \\
X_{ji} &= \text{value of explanatory variable } j \text{ for student } i, \\
\varepsilon_i &= \text{random error term for student } i.
\end{aligned}
$$

The coefficient β_j represents the expected change in academic performance associated with a one-unit change in the corresponding explanatory variable, holding the other explanatory variables constant.

3.4 Model Estimation

The regression coefficients are estimated using the available student-level observations. The fitted model provides estimates of the relationships between the explanatory variables and academic performance and can subsequently be evaluated using appropriate statistical measures (Yang et al., 2018).

3.5 Model Assumptions

The multiple linear regression model is based on assumptions concerning the relationship between the variables and the error term. These include linearity, independence of observations, constant error variance (homoscedasticity), and normality of the errors (Tamhane, 2020).

These assumptions will be considered when evaluating the fitted model to determine whether the regression approach provides an appropriate representation of the observed academic performance data.

3.6 Data Analysis Procedure

The collected data will be organised and analysed using appropriate statistical procedures. Descriptive statistics will first be used to summarise the characteristics of the study variables. The multiple linear regression model will then be fitted to examine the relationship between student academic performance and the selected explanatory variables.

The estimated regression coefficients will be examined to determine the direction and magnitude of the relationships between the explanatory variables and academic performance. Measures of model fit and relevant statistical tests will also be considered when evaluating the fitted model.

The results obtained from these analyses will provide the basis for assessing the research questions and evaluating the usefulness of the proposed statistical model.

4 Results

4.1 Data Description

The analysis is based on a synthetic student-level dataset containing 60 observations. The dataset was constructed for the purpose of this study and contains information on study time, attendance, previous academic performance, access to learning resources, and current academic performance.

Academic performance is treated as the dependent variable, while study time, attendance, previous performance, and access to learning resources are treated as explanatory variables. The dataset therefore provides the variables required to investigate the statistical relationships specified in the methodology.

4.2 Descriptive Statistics

Descriptive statistics were calculated to summarise the distribution of the variables included in the analysis. Table 1 presents the mean, standard deviation, minimum, and maximum values.

Table 1: Descriptive statistics of study variables

Variable	Mean	Standard Deviation	Minimum	Maximum
Study Time	9.15	4.23	2	15
Attendance	84.85	11.17	65	100
Previous Performance	70.33	12.69	46	90
Learning Resources	3.35	1.54	1	5
Academic Performance	57.03	8.42	40.0	76.5

4.3 Exploratory Data Analysis

Exploratory data analysis was conducted to examine the relationships between the explanatory variables and academic performance. The analysis provides an initial assessment of whether differences in study time, attendance, previous academic performance, and access to learning resources are associated with differences in current academic performance.

The patterns observed during this stage provide an initial basis for the subsequent correlation analysis and multiple linear regression model. In particular, the analysis considers whether students with higher levels of study time, attendance, previous performance, and access to learning resources tend to record higher academic performance.

4.4 Correlation Analysis

The Pearson correlation coefficient is used to measure the strength and direction of the linear relationship between academic performance and each explanatory

variable. The correlation coefficient is calculated as

$$r_{XY} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2 \sum_{i=1}^{n}(Y_i - \bar{Y})^2}}. \tag{2}$$

For the 60 student observations, the mean academic performance is

$$\bar{Y} = \frac{\sum Y_i}{60} = 57.0267. \tag{3}$$

The sum of squared deviations of academic performance is

$$\sum_{i=1}^{60}(Y_i - \bar{Y})^2 = 4185.1173. \tag{4}$$

4.4.1 Study Time and Academic Performance

For Study Time, the mean is $\bar{X} = 9.15$. The required components of the correlation calculation are

$$\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 1314.36,$$
$$\sum(X_i - \bar{X})^2 = 1057.65.$$

Therefore,

$$r = \frac{1314.36}{\sqrt{(1057.65)(4185.1173)}} = 0.6247 \approx \mathbf{0.625}. \tag{5}$$

4.4.2 Attendance and Academic Performance

For Attendance, the mean is $\bar{X} = 84.85$. The required components are

$$\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 831.44,$$
$$\sum(X_i - \bar{X})^2 = 7363.65.$$

Therefore,

$$r = \frac{831.44}{\sqrt{(7363.65)(4185.1173)}} = 0.1498 \approx \mathbf{0.150}. \tag{6}$$

4.4.3 Previous Performance and Academic Performance

For Previous Performance, the mean is $\bar{X} = 70.3333$. The required components are

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 3212.3667,$$
$$\sum (X_i - \bar{X})^2 = 9495.3333.$$

Therefore,

$$r = \frac{3212.3667}{\sqrt{(9495.3333)(4185.1173)}} = 0.5096 \approx \mathbf{0.510}. \tag{7}$$

4.4.4 Learning Resources and Academic Performance

For Learning Resources, the mean is $\bar{X} = 3.35$. The required components are

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 83.44,$$
$$\sum (X_i - \bar{X})^2 = 139.65.$$

Therefore,

$$r = \frac{83.44}{\sqrt{(139.65)(4185.1173)}} = 0.1091 \approx \mathbf{0.109}. \tag{8}$$

The resulting correlation coefficients are summarised in Table 2.

Table 2: Correlation between explanatory variables and academic performance

Variable	Correlation with Academic Performance (r)
Study Time	0.625
Attendance	0.150
Previous Performance	0.510
Learning Resources	0.109

4.5 Regression Model Calculation

The multiple linear regression model specified in Equation (1) is applied to the 60 student observations. For clarity, Equation (1) is recalled below:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_p X_{pi} + \varepsilon_i. \tag{1}$$

For this study, the explanatory variables are defined as

$$
\begin{aligned}
X_1 &= \text{Study Time},\\
X_2 &= \text{Attendance},\\
X_3 &= \text{Previous Performance},\\
X_4 &= \text{Learning Resources}.
\end{aligned}
$$

Therefore, Equation (1) becomes

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \varepsilon_i. \tag{9}$$

The regression coefficients are estimated using the ordinary least squares method:

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T Y. \tag{10}$$

For the 60 observations in the dataset, the design matrix X contains a column of ones for the intercept and four columns corresponding to Study Time, Attendance, Previous Performance, and Learning Resources. The required matrix products are

$$X^T X = \begin{bmatrix} 60 & 549 & 5091 & 4220 & 201 \\ 549 & 6081 & 46503 & 38277 & 1758 \\ 5091 & 46503 & 439335 & 358413 & 16856 \\ 4220 & 38277 & 358413 & 306302 & 14349 \\ 201 & 1758 & 16856 & 14349 & 813 \end{bmatrix} \tag{11}$$

and

$$X^T Y = \begin{bmatrix} 3421.6 \\ 32622.0 \\ 291154.2 \\ 243864.9 \\ 11545.8 \end{bmatrix}. \tag{12}$$

Therefore,

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 2.759 \\ 1.452 \\ 0.141 \\ 0.360 \\ 1.096 \end{bmatrix}. \tag{13}$$

Thus, the fitted regression equation is

$$\widehat{Y} = 2.759 + 1.452 X_1 + 0.141 X_2 + 0.360 X_3 + 1.096 X_4. \tag{14}$$

The coefficient of determination, R^2, is calculated as

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \tag{15}$$

where SS_{res} is the residual sum of squares and SS_{tot} is the total sum of squares.

For the fitted model,

$$\begin{aligned} SS_{\text{tot}} &= 4185.1173, \\ SS_{\text{res}} &= 911.3299. \end{aligned}$$

Therefore,

$$R^2 = 1 - \frac{911.3299}{4185.1173} = 0.7822 \approx \mathbf{0.782}. \tag{16}$$

The adjusted coefficient of determination accounts for the number of explanatory variables in the model. It is calculated as

$$\bar{R}^2 = 1 - \left(1 - R^2\right) \frac{n-1}{n-p-1}, \tag{17}$$

where $n = 60$ is the number of observations and $p = 4$ is the number of explanatory variables.

Thus,

$$\bar{R}^2 = 1 - (1 - 0.7822)\frac{59}{55} = 0.7664 \approx \mathbf{0.766}. \tag{18}$$

For example, for a student with 10 hours of study time, 90% attendance, a previous performance score of 70, and a learning-resource score of 4, the predicted academic performance is

$$\begin{aligned} \widehat{Y} &= 2.759 + (1.452)(10) + (0.141)(90) \\ &\quad + (0.360)(70) + (1.096)(4) \\ &= 59.55. \end{aligned}$$

The fitted coefficients and their statistical significance are presented in Table 3.

4.6 Model Diagnostics and Validation

The fitted model was checked by examining the residuals, which represent the differences between the observed and predicted academic performance values. Each residual is calculated as

$$e_i = Y_i - \widehat{Y}_i. \tag{19}$$

Table 3: Multiple linear regression results

Variable	Coefficient	p-value
Intercept	2.759	0.616
Study Time	1.452	< 0.001
Attendance	0.141	0.005
Previous Performance	0.360	< 0.001
Learning Resources	1.096	0.004
R^2	0.782	
Adjusted R^2	0.766	
F-statistic	49.394	< 0.001

Across the 60 observations, the residuals ranged from approximately -9.663 to 9.089, while their mean was approximately zero. The residuals were distributed around zero without a clear systematic pattern.

These checks provide a simple assessment of the fitted model and do not indicate a substantial problem with the model's fit to the synthetic dataset.

4.7 Interpretation of Regression Results

The regression results indicate that all four explanatory variables have positive relationships with academic performance within the fitted model.

Study Time has a coefficient of 1.452 and a p-value of less than 0.001. Holding the other variables constant, a one-unit increase in study time is associated with an estimated 1.452-unit increase in academic performance.

Attendance has a coefficient of 0.141 and a p-value of 0.005. Holding the other variables constant, a one-unit increase in attendance is associated with an estimated 0.141-unit increase in academic performance.

Previous Performance has a coefficient of 0.360 and a p-value of less than 0.001. Holding the other variables constant, a one-unit increase in previous performance is associated with an estimated 0.360-unit increase in academic performance.

Learning Resources has a coefficient of 1.096 and a p-value of 0.004. Holding the other variables constant, a one-unit increase in the learning-resource score is associated with an estimated 1.096-unit increase in academic performance.

All four explanatory variables have p-values below the 5% significance level. This indicates that each variable provides statistically significant evidence of an association with academic performance within the fitted model.

The model has an R^2 value of 0.782, indicating that 78.2% of the variation in academic performance in the synthetic dataset is explained jointly by the four explanatory variables. The adjusted R^2 is 0.766, showing that the model retains substantial explanatory power after accounting for the number of explanatory variables.

Overall, the results suggest that study time, attendance, previous performance, and access to learning resources are useful predictors of academic performance within this synthetic dataset. The fitted model therefore provides a practical statistical framework for examining the factors considered in this study.

5 Conclusion

5.1 Conclusion

This study developed a multiple linear regression model to examine factors associated with student academic performance. Using the 60-observation synthetic dataset, the model considered study time, attendance, previous performance, and access to learning resources as explanatory variables.

The fitted model showed positive relationships between all four explanatory variables and academic performance. Study time, attendance, previous performance, and learning resources were all statistically significant at the 5% level. The model explained approximately 78.2% of the variation in academic performance, while the adjusted R^2 was 76.6%.

Overall, the results demonstrate how multiple linear regression can be used to examine several factors simultaneously and provide a framework for understanding and predicting student academic performance.

5.2 Recommendations

Based on the findings of the study, the following recommendations are made:

1. Students should be encouraged to develop consistent and effective study habits.

2. Educational institutions should promote regular attendance and identify students with attendance difficulties.

3. Previous academic performance should be considered when identifying students who may require additional academic support.

4. Students should be provided with adequate learning resources to support their academic activities.

5. Statistical models can be used as supplementary tools to help educators identify factors associated with student performance.

5.3 Limitations

The dataset used in this study is synthetic and was created for demonstrative purposes. Therefore, the findings should not be interpreted as evidence about a real student population.

In addition, the model considers only the variables included in the dataset. Other factors that may influence academic performance were not examined. The regression results should therefore be viewed as an illustration of the statistical modelling process rather than as definitive conclusions about student academic performance.

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